2024 SSMO Tiebreaker Round

SMO Team

Tiebreaker Round Problem 1: Compute the exact value of

$$2^2 + 0^2 + 2^2 + 4^2 + 20^2 + 22^2 + 24^2 + 40^2 + 42^2 + 202^2.$$

Tiebreaker Round Problem 2: Bob is attempting to make a 3-point shot. Bob attempts the shot 97 times. Each time, Bob has a 35% chance of making the shot. If S_1 denotes Bob's expected score and S_2 the score Bob is most likely to get, then $|S_1 - S_2|$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Tiebreaker Round Problem 3: Let $A = \ldots a_2 a_1 a_0 . a_{-1} a_{-2} a_{-3} \ldots$ be a positive terminating decimal. The length of A is defined to be the length of the shortest sub-sequence of consecutive digits that includes all nonzero digits and at least one of a_0, a_{-1}. So, the length of 12.03 is 4 and the length of 0.123 is 3. Let $f(n)$ be the average of all positive terminating decimals of length n. Find the value of $\left\lfloor \sum_{n=1}^{10} (n+1) f(n) \right\rfloor$.